D18 input inventory

N83 shared-bit encoding · native inventory v2 · 2026-10-21 UTC

Observed inputs and derived counts

The native inventory checks the complete pinned list of 16,627 selected source representatives. It observes 66 common zero coordinates: bit 0 and bits 18–82. Coordinates 1–17 vary. This is an input-list invariant; it does not compute affine rank or establish a general subgroup theorem.

D18 selected roots: shared coordinates

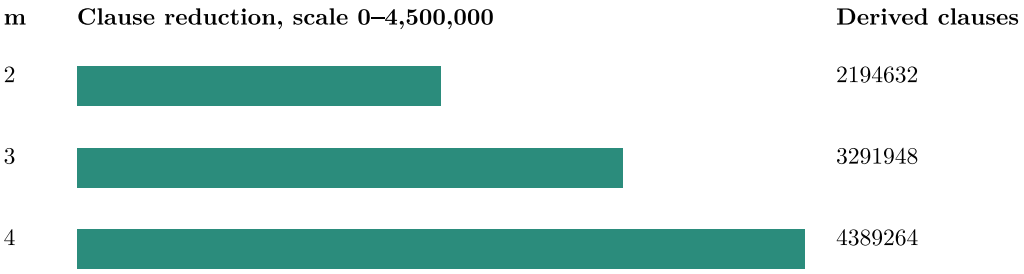
Complete input census: 16627 roots, each with 83 binary coordinates.

Bit 0	Bits 1–17	Bits 18–82
Zero	**Varying**	**Zero**

Coordinate groups are schematic, not proportional to their widths. All 66 shared coordinates are zero. The inventory does not compute affine rank or the dimension of the selected set's span.

Derived root-implication reductions

For m summands: m × 66 × (16627 − 1).



One complete deterministic input census; exact integer counts, no sampling interval. Run: native inventory v2, 2026-10-21 UTC. Source: `root-inventory.json`. Full SAT construction and solver search are separate pending gates.

Reproduction and verification

Curve: `icv1-f2m83-tm6151469093347-debefd74`. The field is GF(2^83), with polynomial z^83+z^45+z^2+z+1. D18 is the source space for the frozen representatives. Their full Frobenius phases and signs remain available; only their pre-phase root coordinates are shared.

Input SHA-256: 534e58ced979e05ec747eb462aa1d63135d8004813d094c522a4d7160c03c4a4. Source SHA-256: 88ac00b0c207c7719c112162604b58143d875d7dbc49a27fa1b5ac7bd5143d5a. The native `root_inventory.rs` uses a u128 OR-of-XOR mask independently of the library-backed construction tool's per-coordinate BigUint check. It checks input digest, header, count, uniqueness, coordinate bounds and positive multiplicities. A changed input and an existing output were both rejected. The wrapper was compiled directly with rustc and already built sha2/serde_json dependencies on the shared Apple arm64 host (L0).

Open gates

No SAT model or solver search is executed by this inventory. The chart's reductions are derived from m × 66 × 16,626; total model counts require the full construction check. The checked Lean theorem proves the Boolean replacement and count identity under its stated hypotheses; it does not verify this parser, compiler or complete solver. Paired controls, natural query yield/useful rank, cold/reused pipeline cost and matched rho remain open. There is no runtime comparison or selected candidate in this receipt.

Sources and exact commands: `INVENTORY.md`, `root-inventory.json`, `root_inventory.rs`, `PROOF_MAP.md`, `lean/SharedRootBits.lean`, and `PROTOCOL.md`. Canonical performance dashboards are unchanged because this result supplies no measured time or pipeline ranking.